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Productivity Impairment Caused by Placement of Micro-Sized Proppants in Micro-Fractures: How

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Abstract

Micro-sized proppants are increasingly applied in hydraulic fracturing to prop micro-fractures and expand the stimulated reservoir volume (SRV) in unconventional formations. Although some oilfields have observed production gains from their use, others have encountered well performance deterioration. Given that the design of micro-sized proppant injection remains largely empirical, the mechanisms underlying reservoir damage caused by these proppants are not fully understood. This uncertainty has discouraged their broader adoption in some fields. To address this, a model was developed using the linear elastic deformation theory for both formation rock and proppant particles, quantifying conductivity changes in micro-fractures containing a partial monolayer of these proppants. By comparing the predicted propped fracture conductivity with its initial value, this study highlights when proppant placement enhances or reduces conductivity, categorizes damage mechanisms, and introduces a conceptual framework distinguishing "improvement zones" from "damage zones." A propped fracture productivity model was developed by integrating the conductivity model, resulting in a diagram that correlates folds of productivity increase with stimulation distance.

Modeling results indicate that the fold increase in well productivity does not monotonically rise with the increasing placement distance; instead, there is a turning point beyond which further placement reduces productivity gains. A case study of micro-sized proppant stimulation from a Coalbed Methane (CBM) reservoir in the Qinshui Basin, China demonstrates that as the dimensionless stimulation radius increases from 0 to 0.86, the folds of productivity increase rise from 1.0 to 5.9. However, when the radius exceeds 0.86, productivity gains begin to decline. This suggests that the optimal stimulation radius is approximately 86% of the drainage radius; stimulating an entire drainage zone does not yield the highest productivity. The underlying mechanism is that micro-fractures farther from a wellbore experience lower closure pressures during production, so the proppants' contribution to fracture opening is outweighed by their detrimental impact on micro-fracture permeability. However, notably, the fold increase in productivity rises from 1.6 to 4.7 as the injection rate of proppant-carrying fluid increases from 1.0 m^3/min to 10.0 m^3/min. A higher fluid injection rate always leads to better stimulation effectiveness. This work clarifies the mechanisms

behind productivity impairment, emphasizing both the risks and opportunities of micro-sized proppant use. By identifying optimal design parameters, it provides practical guidance to improve stimulation strategies and maximize unconventional reservoir development.
and When it Occurs

Introduction

Hydraulic fracturing with proppants is a widely used technique to stimulate reservoirs and achieve economic production rates (Rahman et al., 2007; Asadi et al., 2020; Aslannezhad et al., 2021). However, many natural fractures or secondary artificial fractures—especially those farther from a wellbore—have apertures too small to accommodate larger proppant particles. As a result, only the regions near the wellbore with relatively larger fracture apertures are effectively stimulated (Dejam, 2019). To address this limitation, researchers have proposed a method known as Graded Proppant Injection (GPI), which involves sequentially injecting smaller proppant particles followed by larger ones (Bedrikovetsky et al., 2012; Khanna et al., 2013). The smaller particles can penetrate deeper into a formation, while larger particles tend to be trapped near a wellbore, leading to more uniform proppant distributions throughout a fracture system. Given the extremely narrow kinematic apertures of micro fractures (Gale et al., 2014), the use of micro-sized proppants to prop them is essential (Lau et al., 2019). Moreover, in such conditions, a monolayer of proppants is typically placed within a fracture system, as the secondary and tertiary fractures are too narrow to accommodate multiple proppant packs (Warpinski et al., 2009). Thus, GPI typically involves the monolayer placement of micro-sized proppants (Keshavarz et al., 2016; Khanna et al., 2013). Furthermore, a partial monolayer, rather than a densely packed monolayer, is a typical placement pattern to maximize fracture conductivity in narrow fractures.

Unlike the placement of multilayer proppant packs in hydraulic fractures, the placement of monolayer proppants in micro fractures requires careful optimization of the proppant concentration (or a proppant packing ratio, defined as the ratio of a proppant particle diameter to the distance between particles). The proppant packing ratio should neither be too high nor too low, as two competing mechanisms influence fracture conductivity: (1) deformation of flow channels due to sparse proppant placement and (2) increased tortuosity of the flow channels due to dense proppant placement (Khanna et al., 2013). To date, there is no quantitative design method for micro-sized proppant injection, despite attempts to obtain some parameters quantitatively such as propped fracture permeability in the presence of micro-proppants in micro-fractures (Kim et al., 2018) and, most importantly, Optimal Proppant Packing Ratio (OPPR). The investigation of OPPR has been conducted by Khanna et al. (Khanna et al., 2012, 2013) and Keshavarz et al. (Alireza Keshavarz et al., 2015; Keshavarz et al., 2016), who all used a method of plotting a diagram of a permeability correction factor against a proppant packing ratio. They also conducted core-flood tests to validate their findings. An OPPR calculation model has been integrated into a GPI model to determine an optimal injection schedule and predict productivity increases (Khanna et al., 2013; A. Keshavarz et al., 2015; Keshavarz et al., 2016). However, in their models, proppant particles were assumed to be rigid, and the deformation and embedment of proppants into a fracture wall were neglected. Liu et al. (Liu et al., 2020a, 2020b) takes Proppant Embedment and Proppant Deformation (PEPD) into consideration in their models; however, the initial fracture width is assumed to be equal to the proppant particle diameter, in which case the Permeability Correction Factor (PCF) as well as the Conductivity Correction Factor (CCF) are always smaller than 1.0. This cannot indicate the potential damage caused by placement of micro-sized proppants.

In terms of field applications, there are some published data showing the positive effects of use of micro-sized proppants in well stimulation in the past decade (Dahl et al., 2015a, 2015b; Ji et al., 2016; Madasu and Nguyen, 2017; Li et al., 2018; Lau et al., 2019). For instance, Dahl et al. (Dahl et al., 2015a, 2015b) introduced the use of 30 μm micro-sized proppants to enhance well production in the Barnett shale. Li et al. (Kim et al. 2018) conducted a pilot project using small-sized proppants in major unconventional formations in the Haynesville shale, demonstrating positive post-fracturing production performance (Li et al. 2018). Ji et al. presented a case study in the Long Ma Xi Shale, China, where the use of micro-proppants led to an improved SRV (Stimulated Reservoir Volume) and increased post-stimulation production (Ji et al. 2016). Nevertheless, there is also some feedback from oilfields, even though not published, indicating a production rate decease after conducting proppant fracturing. The mechanism of the damage remains to be figured out and the phenomenon needs to be quantitatively analyzed.

This study investigates the mechanisms of productivity impairment caused by the placement of micro-sized proppants in micro- fractures and quantifies the conditions under which such damage occurs. While many studies emphasize the benefits of micro-sized proppants, this work raises awareness of their potential negative effects and offers insights into optimizing stimulation designs to mitigate these risks. These findings contribute to more effective well stimulation strategies for the development of unconventional reservoirs.

Propped Fracture Conductivity

When proppant suspension is injected into a fracture via surface pumping, two competing mechanisms come into play. First, the hydraulic energy dilates the fracture aperture, and after the pumping pressure is relieved, the proppant particles prop the fracture open, creating a wider flow path. Second, the proppant particles trapped within the fracture introduce resistance to fluid flow. Whether an injection enhances productivity depends on which mechanism dominates. In some cases, despite the reduction in fracture permeability, a wider fracture aperture was achieved. The combined effect of these two mechanisms, reduced permeability and increased fracture width, resulted in an overall conductivity increase. However, in other cases, the permeability reduction outweighs the aperture increase, resulting in impaired fracture conductivity and reduced well productivity.

Basic Models

The propped fracture conductivity, $C_f = k_f w_f$, is introduced to analyze the impact of micro-sized proppant placement, where k_f denotes the propped fracture permeability and w_f is the propped fracture width. k_f can be estimated by obtaining an equivalent value within the propped area, which can be expressed as follows (Liu et al., 2020a)

$$k_f = k_{eq}^{1,2,3} = \frac{k_1 k_2}{l_d k_2 / D_1 - k_2 + k_1} + \frac{k_3\left(l_d / D_1 - 1\right)}{l_d / D_1} \tag{1}$$

where $k_{eq}^{1,2,3}$ is the equivalent permeability of Area 1, Area 2 and Area 3 shown in Fig.1. D_1 is proppant particle diameter. l_d is the distance between proppant particles. k_1, k_2, k_3 are the permeability of Area 1, Area 2 and Area 3, respectively, which can be expressed as follows

$$k_1 = k_3 = \frac{w_f^2}{12} \tag{2}$$

$$k_2 = \frac{\phi_2^3}{36 C_k\left(1-\phi_2^2\right)} D_1^2 \tag{3}$$

$$\phi_2 = \frac{D_1 - 2\gamma - \frac{\pi}{6}\left(D_1 - 2\zeta\right) + \frac{2\pi}{3}\left(\frac{3}{2}D_1 - h\right)\frac{h^2}{D_1^2}}{D_1 - 2\gamma} \tag{4}$$

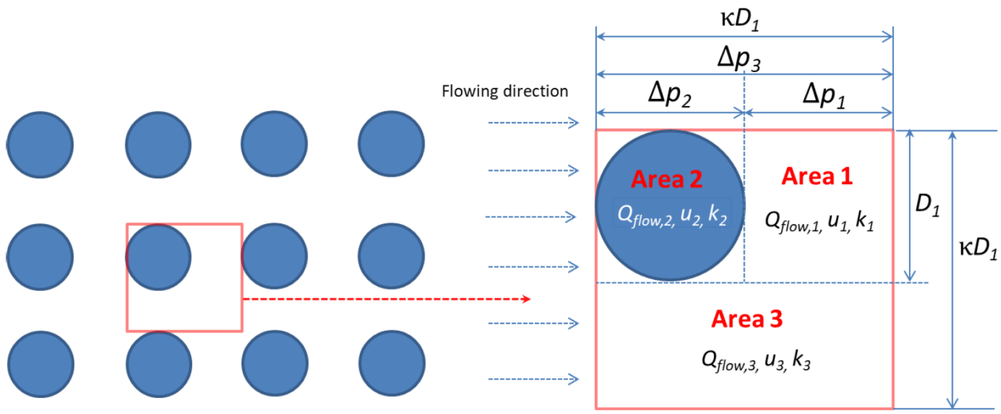

Figure 1—A symmetry element for calculating the equivalent fracture permeability (Liu et al., 2020a)

In Eqs. (3) and (4), C_k is the Carman-Kozeny constant and equals 5 for spherical particles. ϕ_2 is the porosity of Area 2. ζ represents the proppant deformation. h is the proppant embedment. γ is the change in fracture aperture in one rock bed (half the total change in fracture aperture). ζ, h and γ are calculated by an analytical model derived by Kewen Li et al.(Li et al., 2015). Considering the effect of Proppant Embedment and Proppant Deformation (PEPD), the fracture width, w_f, is expressed as $w_f = w_{f0} - 2\gamma$, where w_{f0} is the initial fracture width.

Conductivity Correction Factor (CCF) is introduced and defined as the ratio of the propped fracture conductivity to the initial fracture conductivity:

$$CCF = \frac{C_f(\beta_p, \varepsilon_\sigma)}{C_{f0}} = \frac{k_{eq}^{1,2,3}(\beta_p, \varepsilon_\sigma) \cdot w_f(\beta_p, \varepsilon_\sigma)}{k_{f0} \cdot w_{f0}} \tag{5}$$

where C_{f0} is the initial fracture conductivity, defined as: $C_{f0} = k_{f0}w_{f0}$. Here, the initial fracture width w_{f0} is the fracture aperture at reservoir pressure. It is difficult to measure directly but can be deduced from mechanical calculations or fracture reconstruction using cores. k_{f0} is the initial permeability of a natural fracture, meaning the fracture permeability at reservoir pressure. It can be estimated by Eq. (2). β_p is the proppant packing ratio and $\beta_p = D_1/l_d$. ε_σ is the dimensionless effective stress. $k_{eq}^{1,2,3}$ and w_{f0} are functions of β_p and ε_σ because the propped fracture permeability and width vary with the density of proppant packing and the stress exerted on the fracture surface. The overall effect is that CCF is a function of β_p and ε_σ.

Given the conditions: initial fracture width w_{f0} =0.5mm, proppant diameter D_1 = 1.0mm, proppant elastic modulus E_1 = 10000MPa, rock elastic modulus E_2 = 4000MPa, Poisson's ratios of the proppant and the rock are $v_1 = v_2 = 0.2$, and thicknesses of the upper and lower rock beds that deform when closure pressure is applied D_2 = 3.0mm, a diagram of CCF vs. proppant packing ratio at different levels of dimensionless effective stress is plotted, which is shown in Figure 2.

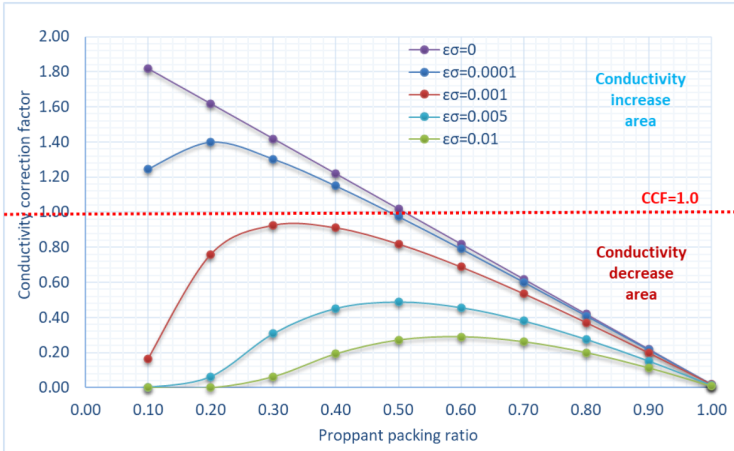

Figure 2—CCF vs. proppant packing ratio at different values of dimensionless effective stress

Damage Zones

This model above allows for evaluating whether a specific micro-sized proppant placement scheme will have positive or negative effects on fracture conductivity. In Figure 2, the area above the red dotted line (CCF > 1) represents the "conductivity increase area", meaning the schemes with positive effects on fracture conductivity. Conversely, the area below the red dotted line (CCF≤1) is the "conductivity decrease area", indicating schemes that damage the fracture conductivity rather than improving it, even if Optimal Proppant Packing Ratio (OPPR) is achieved. OPPR is the value of the packing ratio that makes the maximum CCF, which is indicated at the peak of the curves in Figure 2.

Given a fracture with an initial width, OPPR at a specific closure stress can be obtained by a delicate proppant placement design, then the corresponding CCF can be obtained from Fig. 2. Therefore, what we care most is actually the OPPR and the CCF at OPPR, namely the peak points in the curves in Fig. 2. If the calculated dimensionless effective stress does not fall in the existing curves, method of interpolation can be used to get the intermediate value. If we extract the OPPR points from the CCF vs. proppant packing ratio curves, a trendline can be drawn to show the variation of CCF $\left(\beta_p^*, \varepsilon_\sigma\right)$ with OPPR, which is illustrated in Fig. 3. We can call this trendline as a "OPPR line".

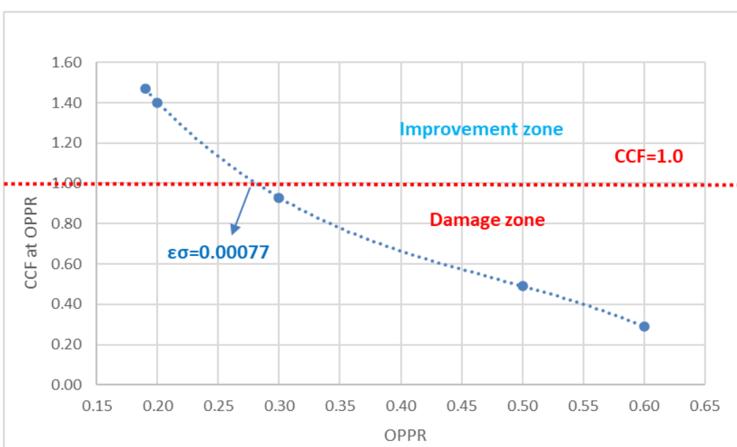

Figure 3—CCF when proppants are placed at OPPR

From Fig. 3 we can see that the line of CCF=1.0 divide the chart into two zones: improvement zone for CCF≥1.0 and damage zone for CCF<1.0. Similarly, the OPPR line is cut into two sections: improvement section and damage section. In a given case, the closure stress on fracture during production plays an important role in whether a proppant placement scheme at OPPR will improve the fracture conductivity or damage it. In this given case, when the calculated dimensionless effective stress is smaller than 0.00077, a proppant placement scheme at OPPR can improve the propped fracture conductivity; otherwise, it damages the propped fracture conductivity. When designing a proppant placement scheme, we must make sure that our designed CCF falls into the improvement zone.

However, if the initial fracture width is too large, no matter how we place the proppants and no matter how small the effective stress is, the CCF will always fall in the damage zone. If we change the initial fracture width to 1.0mm and keep all the other input data the same for this case, the CCF vs. proppant packing ratio curves are drawn and shown in Fig. 4. Obviously, all the points and curves fall into the damage zone where CCF<1.0. This indicates that micro-sized proppant placement is not applicable in such a well. No matter how well you design your proppant placement scheme, it will damage the propped fracture conductivity instead of improving it.

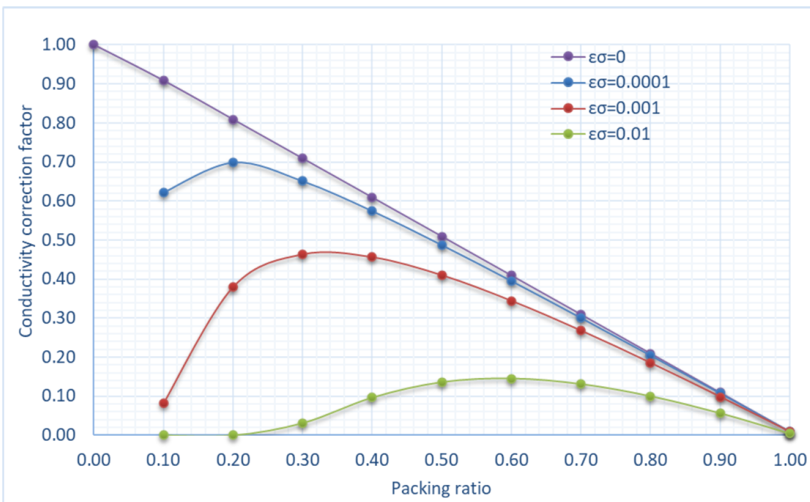

Figure 4—CCF vs. proppant packing ratio at different values of dimensionless effective stress (W_{f0}=1.0 mm)

On the other hand, if the initial aperture of fractures is very small, the proppant placement design results will fall into the improvement zone more easily. For the same case, if we change the initial fracture width to 0.05mm and keep all the other input data the same, the CCF vs. proppant packing ratio curves are shown in Fig. 5. We can see that all the OPPR points for the six curves are in the improvement zone. Even when the dimensionless effective stress is as large as 0.01 (the green curve), the propped fracture conductivity is tripled if we place the proppants at OPPR=0.60. This indicates that the partial monolayer micro-sized proppant placement is especially applicable when the initial aperture of fractures is small.

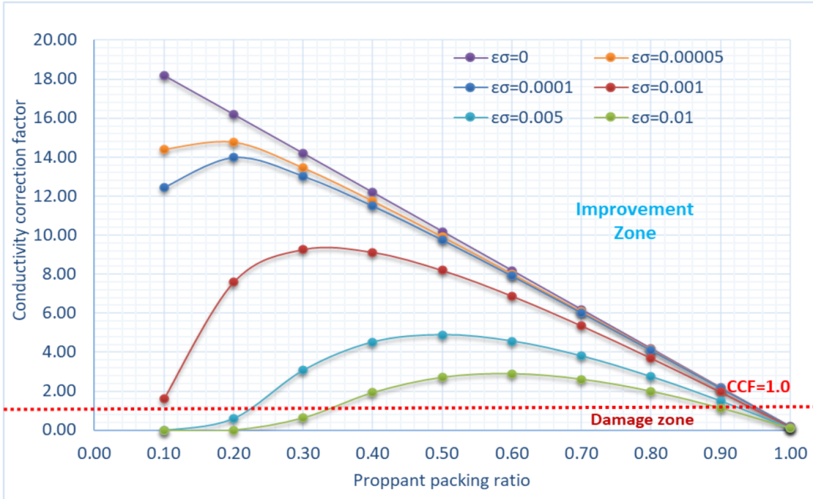

Figure 5—CCF vs. proppant packing ratio at different values of dimensionless effective stress (w_{f0} = 0.05 mm)

Propped Fracture Productivity

The above analysis assumes uniform openings (or widths) of natural fractures. However, even if the openings of natural fractures are uniform under original reservoir conditions, they will be dilated by hydraulic energy during injection of fracturing fluid. Because the hydraulic energy is strong near the wellbore and is gradually weakened due to leakage, the dynamic fracture during injection should be wedgy. Therefore, after placement of micro-sized proppants under GPI design scheme, the propped fracture will also be wedgy, as is shown in Fig. 6.

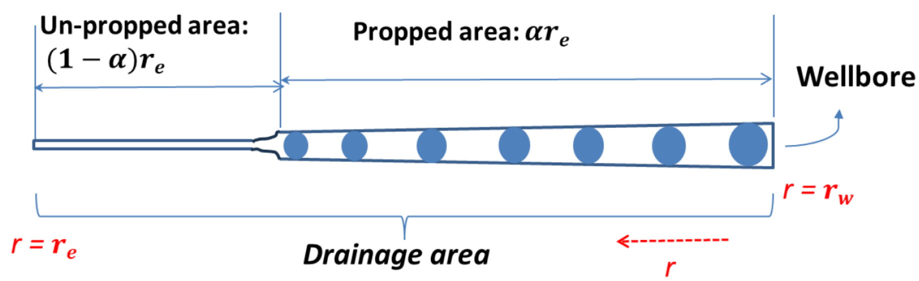

Figure 6—Wedgy propped fracture shape after GPI treatment

Since the propped fracture opening is not uniform, we cannot directly evaluate if a GPI design fall into the "improvement zone" or "damage zone" through the CCF vs. proppant packing ration diagram. The effect of a GPI treatment is improvement of conductivity in one fracture section but damage of conductivity in another. In the near-wellbore region, natural fractures are dilated, and their conductivity is likely enhanced due to the strong hydraulic effect (though they may still be damaged if the expansion of fracture opening fails to offset proppant-induced plugging). In the area that is far from the wellbore, the fracture conductivity is probably damaged if placed with proppants, because the hydraulic energy is weakened and the fracture opening dilation is very limited, which cannot counteract the permeability reduction caused by proppant placement.

Therefore, the effect of a GPI scheme on fracture productivity is a comprehensive outcome, and it may be impacted by the placement distance and the injection rate of fracturing fluid. When designing a GPI scheme, it is necessary to rapidly predict the resulting productivity increase to evaluate the overall efficiency of the proposed proppant placement plan.

Basic Models

Given the context of Graded Proppant Injection (GPI), a model for predicting the post-stimulation productivity improvements for a naturally fractured formation can be established. Khanna et al. derived an analytical formula (Khanna et al., 2013) to calculate the "folds of increase" in productivity for naturally fractured reservoirs due to graded proppants placement. Based on our study results, we have modified this formula to incorporate OPPR while considering Proppant Embedment and Proppant Deformation (PEPD) and using CCF' instead of Permeability Correction Factor. The updated formula for predicting productivity increase is expressed as follows:

$$\eta = \frac{CCF'\left(\beta_p^*, \varepsilon_\sigma\right)\varepsilon_q \ln\left(r_e / r_w\right)}{\frac{4\left[w_f\left(r_w\right) - w_f\left(r_e\alpha\right)\right]}{w_{f0}} - CCF'\left(\beta_p^*, \varepsilon_\sigma\right)\varepsilon_q \ln\alpha} \tag{6}$$

where η represents the folds of increase in the productivity index; CCF' $\left(\beta_p^*, \varepsilon_\sigma\right)$ is the conductivity correction factor at OPPR under dimensionless effective stress ε_σ. The only difference between CCF' and CCF is that the initial fracture width for CCF' is assumed as the fracture width when $\varepsilon_\sigma = 0$ and the fracture is empty, namely, the largest fracture width during GPI. Therefore, CCF' is always smaller than 1.0. ε_q is the dimensionless injection rate, which is defined as $\varepsilon_q = 2q_{in}\mu_{fr}C_r/(\pi k_{f0}\phi)$, where μ_{fr} is the viscosity of fracturing fluid, C_r is the reservoir compressibility under uniaxial strain, k_{f0} is the initial fracture permeability, and φ is the reservoir porosity. β_p^* is the Optimal Proppant Packing Ratio (OPPR), which can be obtained from the CCF vs. packing ratio diagram in Figure 2. Two steps can be taken to determine β_p^*: Firstly, perform a polynomial fit of the CCF vs. packing ratio curve to obtain its polynomial function. Then, calculate its derivative function to find the extreme value of CCF. The extreme value of CCF is CCF $\left(\beta_p^*, \varepsilon_\sigma\right)$, and the packing ratio at this extreme value is β_p^*.

The fracture apertures $w_f(r_w)$ at the wellbore and $w_f\,(r_e a)$ at the stimulation radius are calculated from the following equations:

$$w_f\left(r_w\right) = w_{f0}\left[1 + \varepsilon_q \ln\left(r_e / r_w\right)\right]^{1/4} \tag{7}$$

$$w_f\left(r_e\alpha\right) = w_{f0}\left[1 - \varepsilon_q \ln\alpha\right]^{1/4} \tag{8}$$

Effect of Stimulation Radius

We then applied the model to a real case from a coalbed methane (CBM) formation in the Qinshui Basin, Shanxi, China, to analyze the effects of GPI stimulation. The basic input data are listed in Table 1.

Table 1—Input data for GPI stimulation analysis in Qinshui Basin

Parameters	Value	Parameters	Value
Formation thickness	$H = 10$m	Proppant Young modulus	E_1 =10000MPa
Initial fracture opening	$W_{f0} = 20$μm	Formation rock Young modulus	E_2 =1250MPa
Spacing of cleats	$l_c = 0.05$m	Proppant Poisson's ratio	$v_1 = 0.20$
Drainage radius	$r_e = 500$m	Formation rock Poisson's ratio	$v_2 = 0.33$
Fracture closure pressure	p = 4MPa	Injection rate	$q_{In} = 10.0$m^3/min
Calculated dimensionless effective stress	ε_σ =0.00285	Calculated dimensionless injection rate	$E_q = 45.8$
Expected dimensionless stimulation radius: a = 0.4			

The OPPR for this case is obtained by plotting the CCF vs. packing ratio diagram, as shown in Figure 7. The result is OPPR=0.41, and the corresponding CCF' $\left(\beta_p^*, \varepsilon_\sigma\right)$. Submitting this data together with that data in Table 1 into Eqs. (6)–(8) yields the fold of productivity increase caused by the partial-monolayer placement of proppants with a packing ratio of 0.41 is 4.68.

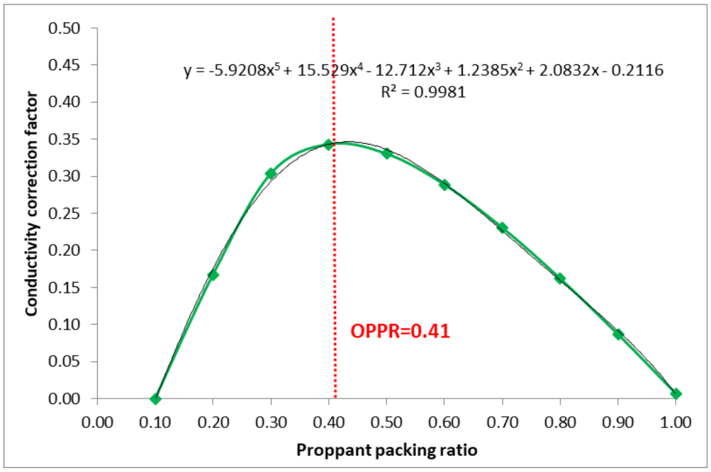

Figure 7—CCF vs. packing ratio diagram calculated from the data in Table 1

By varying the dimensionless stimulation radius while keeping all other parameters constant, we obtain the curve of fold of productivity increase versus scaled stimulation radius, as shown below.

The results shown in Fig. 8 indicate that GPI stimulation effectiveness generally improves as the stimulation radius increases, with the fold of productivity increase reaching a maximum of 5.9 under the given conditions. However, when the scaled stimulation radius exceeds 0.86, productivity gains begin to decline. This decline occurs because, in this case, the natural fractures already have an initial opening of 20 μm Under such circumstances, proppant placement may fail to enhance fracture conductivity and can even cause damage—a phenomenon also observed in Fig. 2. These findings highlight the importance of carefully selecting the stimulation radius rather than stimulating the entire drainage area indiscriminately. Choosing an optimal stimulation radius is critical to maximizing the effectiveness of GPI treatment.

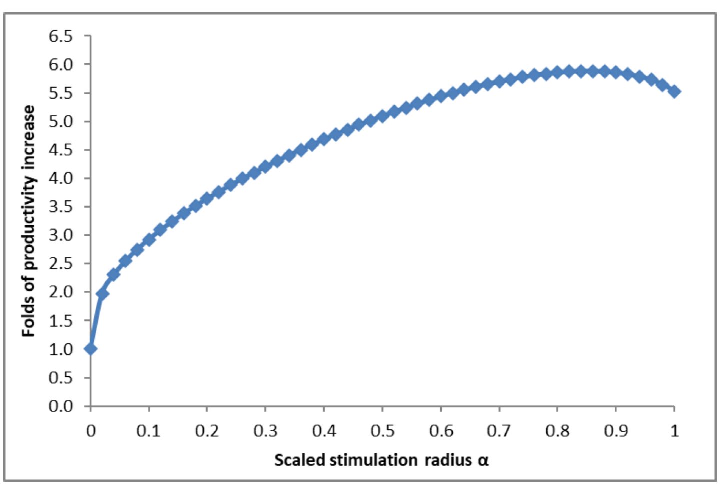

Figure 8—Folds of productivity increase with different stimulation radii (injection rate=10.0m^3/min)

By varying the injection rate, optimal scaled stimulation radii that maximize the fold of productivity increase can still be identified, as shown in Fig. 9. The results also reveal that lower injection rates correspond to smaller optimal scaled stimulation radii. This implies that when the allowable injection rate on-site is limited, a smaller stimulation radius is sufficient, and extending the stimulation distance further may have detrimental effects. Therefore, accurately determining the optimal stimulation radius is especially crucial in GPI design.

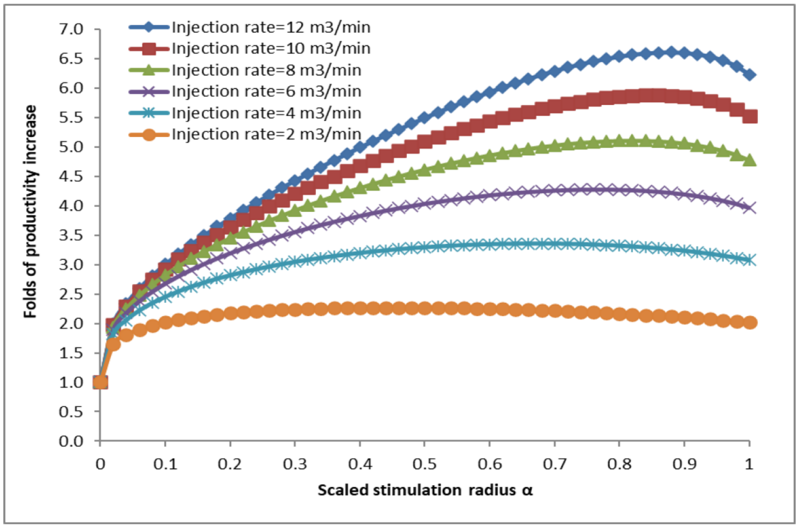

Figure 9—Change of folds of productivity increase with scaled stimulation radii at different injection rates

For the case presented in Table 1, the optimal scaled stimulation radius (a*) at various injection rates can be extracted from Fig. 9 and is more directly illustrated in Fig. 10.

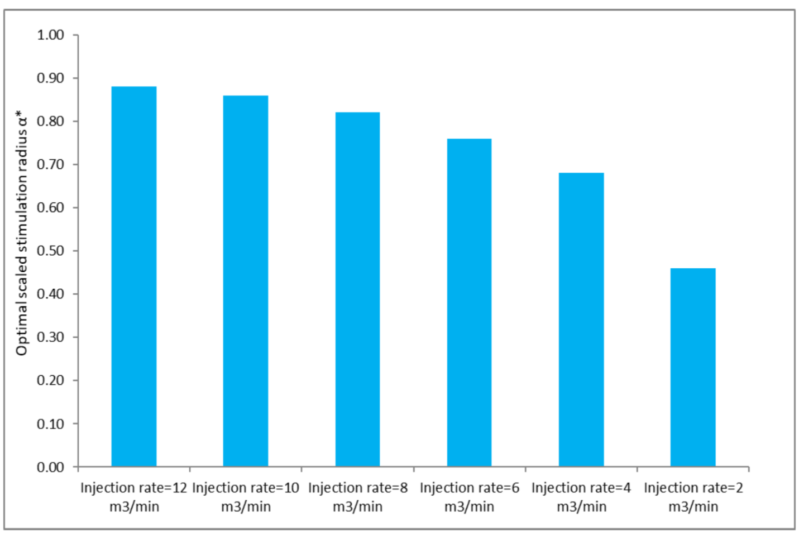

Figure 10—The optimal scaled stimulation radius at different injection rates

Effect of Injection Rate

To investigate the effect of injection rate on the fold of productivity increase, we modified the injection rate in Table 1 from 1.0 m^3/min to 10.0 m^3/min, while keeping all other parameters unchanged. The results, shown in Fig. 11 ($a = 0.4$), indicate that GPI stimulation effectiveness consistently improves with increasing injection rate. This is expected, as a higher injection rate produces a larger dynamic fracture width, regardless of the initial fracture width. Therefore, when designing a GPI treatment, it is advisable to set the injection rate as high as operationally feasible to maximize post-stimulation productivity gains.

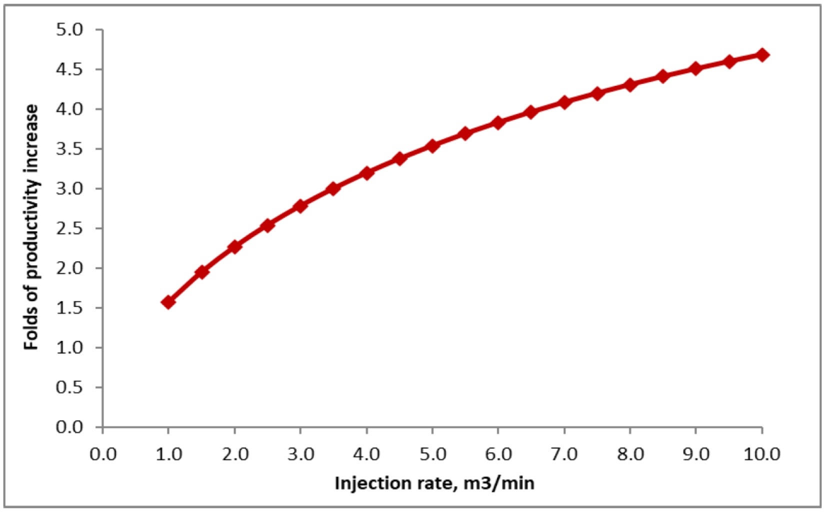

Figure 11—Folds of productivity increase with different injection rate (α = 0.40)

To confirm that this trend is consistent under different conditions, the scaled stimulation radius was reduced to $\alpha = 0.2$ and increased to $\alpha = 0.8$. The corresponding results, presented in Fig. 12, show that the fold of productivity increase rises with injection rate in all cases, reinforcing the conclusion that higher injection rates enhance GPI performance.

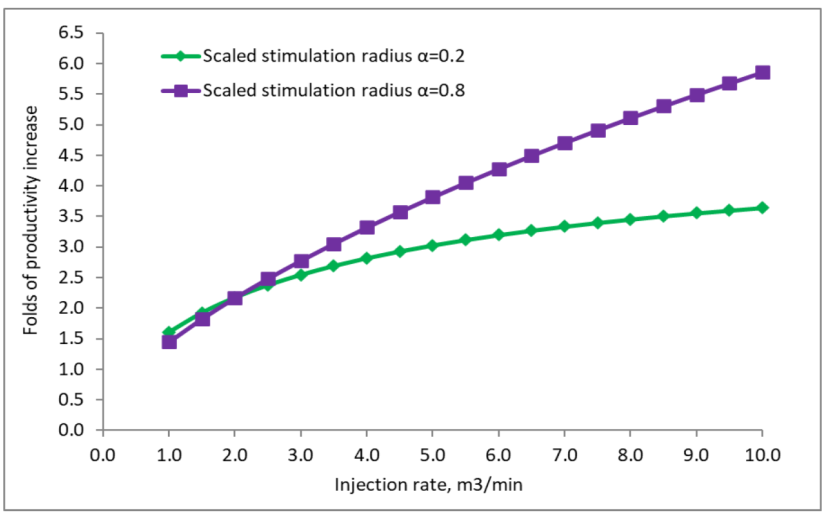

Figure 12—Folds of productivity increase with different injection rate (α = 0.20 and α = 0.80)

A Recommended Workflow

Based on the above analysis, we recommend first constructing the CCF-Proppant Packing Ratio diagram to assess the applicability of GPI stimulation. GPI design should only proceed if the Optimal Proppant Packing Ratio (OPPR) point lies within the "improvement zone." Once applicability is confirmed, the Fold of Productivity Increase-Scaled Stimulation Radius diagram should be generated to determine the Optimal Scaled Stimulation Radius. This parameter serves as a critical input for GPI design, ensuring the maximum possible increase in fracture productivity.

The recommended workflow is illustrated in Fig. 13.

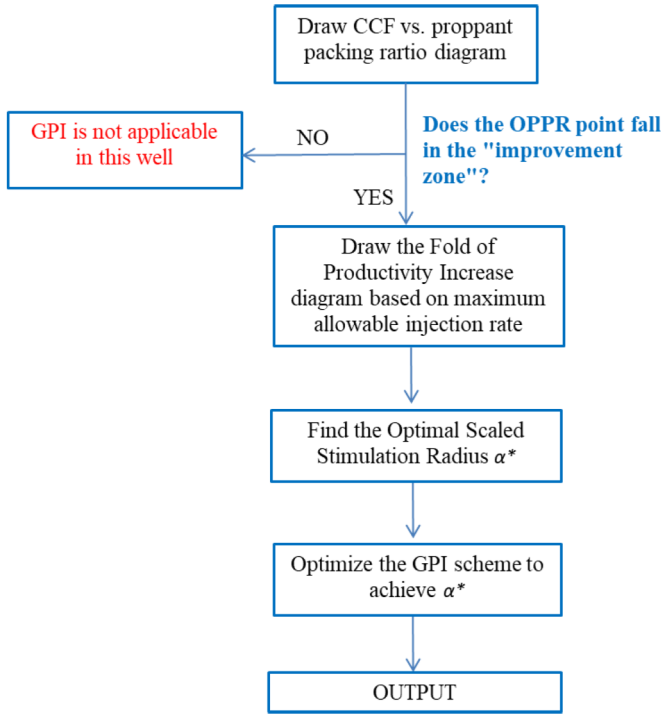

Figure 13—A recommended workflow to optimize GPI design

Conclusion

This study investigates the mechanisms of productivity impairment resulting from the placement of micro-sized proppants in micro- fractures and quantifies the conditions under which such damage occurs, using a field case from the Qinshui Basin, China. Based on the analysis, the following conclusions are drawn:

1. In partial monolayer GPI stimulation, the use of micro-sized proppants can impair fracture conductivity, particularly when the initial fracture opening is large. This occurs because the permeability reduction from proppant plugging outweighs the fracture-width dilation provided by proppant support.
2. A diagram correlating CCF with proppant packing ratio has been developed, enabling estimation of the positive or negative impact of micro-sized proppant placement on propped fracture conductivity. This tool can be applied to screen the suitability of GPI for specific wells.
3. A GPI-specific propped fracture productivity prediction model has been established. Using this model, a diagram of Fold of Productivity Increase versus Scaled Stimulation Radius can be generated, allowing determination of the optimal scaled stimulation radius.
4. Lower injection rates of proppant-carrying fluid result in smaller optimal scaled stimulation radii. Where feasible, injection rates should be maximized to enhance stimulation effectiveness.
5. A workflow is proposed for pre-screening candidate wells and optimizing GPI design, with the objective of maximizing post- stimulation productivity.

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Acronyms

CCF Conductivity Correction Factor
PEPD Proppant Embedment and Proppant Deformation
OPPR Optimal Proppant Packing Ratio
CBM Coal Bed Methane

References

Asadi, M.B., Dejam, M., Zendehboudi, S., 2020. Semi-analytical solution for productivity evaluation of a multi-fractured horizontal well in a bounded dual-porosity reservoir. *Journal of Hydrology* **581**, 124288. https://doi.org/10.1016/j.jhydrol.2019.124288

Aslannezhad, M., Kalantariasl, A., You, Z., Iglauer, S., Keshavarz, A., 2021. Micro-proppant placement in hydraulic and natural fracture stimulation in unconventional reservoirs: A review. *Energy Reports* **7**, 8997–9022. https://doi.org/10.1016/j.egyr.2021.11.220

Bedrikovetsky, P., Keshavarz, A., Khanna, A., Kenzie, K.M., Kotousov, A., 2012. Stimulation of Natural Cleats for Gas Production From Coal Beds by Graded Proppant Injection. Presented at the SPE Asia Pacific Oil and Gas Conference and Exhibition, OnePetro. https://doi.org/10.2118/158761-MS

Dahl, J., Calvin, J., Siddiqui, S., Nguyen, P., Dusterhoft, R., Holderby, E., Johnson, B., 2015a. Application of micro-proppant in liquids-rich, unconventional reservoirs to improve well production: Laboratory results, field results, and numerical simulations, in: Abu Dhabi International Petroleum Exhibition and Conference. OnePetro.

Dahl, J., Nguyen, P., Dusterhoft, R., Calvin, J., Siddiqui, S., 2015b. Application of micro-proppant to enhance well production in unconventional reservoirs: Laboratory and field results, in: SPE Western Regional Meeting. OnePetro.

Dejam, M., 2019. Tracer dispersion in a hydraulic fracture with porous walls. *Chemical Engineering Research and Design* **150**, 169–178. https://doi.org/10.1016/j.cherd.2019.07.027

Gale, J.F., Laubach, S.E., Olson, J.E., Eichhubl, P., Fall, A., 2014. Natural fractures in shale: A review and new observations. *AAPG bulletin* **98**, 2165–2216.

Ji, P., Qu, Q., Vitthal, S., Pickles, M., 2016. Improving propped SRV in the long Ma Xi Shale: New concept trials and its performance, in: SPE Asia Pacific Hydraulic Fracturing Conference. OnePetro.

Keshavarz, Alireza, Badalyan, A., Carageorgos, T., Bedrikovetsky, P., Johnson, R., 2015. Graded Proppant Injection into Coal Seam Gas and Shale Gas Reservoirs for Well Stimulation. Presented at the SPE European Formation Damage Conference and Exhibition, OnePetro. https://doi.org/10.2118/174200-MS

Keshavarz, A., Badalyan, A., Johnson, R., Bedrikovetsky, P., 2016. Productivity enhancement by stimulation of natural fractures around a hydraulic fracture using micro-sized proppant placement. *Journal of Natural Gas Science and Engineering* **33**, 1010–1024. https://doi.org/10.1016/j.jngse.2016.03.065

Keshavarz, A., Badalyan, A., Johnson, R., Bedrikovetsky, P., 2015. Stimulation of Unconventional Reservoirs using Graded Proppant Injection. Presented at the SPE Russian Petroleum Technology Conference, OnePetro. https://doi.org/10.2118/176712-MS

Khanna, A., Keshavarz, A., Mobbs, K., Davis, M., Bedrikovetsky, P., 2013. Stimulation of the natural fracture system by graded proppant injection. *Journal of Petroleum Science and Engineering* **111**, 71–77. https://doi.org/10.1016/j.petrol.2013.07.004

Khanna, A., Kotousov, A., Sobey, J., Weller, P., 2012. Optimal placement of proppant in narrow fracture channels. J. Pet. Sci. *Eng* **100**, 9–13.

Kim, B.Y., Akkutlu, I.Y., Martysevich, V., Dusterhoft, R., 2018. Laboratory measurement of microproppant placement quality using split core plug permeability under stress, in: SPE Hydraulic Fracturing Technology Conference and Exhibition. OnePetro.

Lau, H.C., Radhamani, A.V., Ramakrishna, S., 2019. Maximizing Production from Shale Reservoir by Using Micro-Sized Proppants. Presented at the International Petroleum Technology Conference, OnePetro. https://doi.org/10.2523/IPTC-19437-MS

Li, C., Spurr, N., Royce, T.N., 2018. Post-fracturing production performance of small sized proppant in major unconventional formations, in: SPE International Hydraulic Fracturing Technology Conference and Exhibition. OnePetro.

Li, K., Gao, Y., Lyu, Y., Wang, M., 2015. New Mathematical Models for Calculating Proppant Embedment and Fracture Conductivity. *SPE Journal* **20**, 496–507. https://doi.org/10.2118/155954-PA

Liu, H., Li, J., Xu, Z., Yuan, Z., Liu, J., Ren, D., Feng, J., 2020a. Influences of proppant embedment and deformation on micro-sized graded proppant injection scheme, in: SPE Asia Pacific Oil and Gas Conference and Exhibition. SPE, p. D013S103R020.

Liu, H., Li, J., Yuan, Z., Liu, J., Ren, D., Feng, J., Lv, Y., 2020b. Modelling of micro-sized graded proppant injection in the natural fractures around a hydraulic fracture with consideration of proppant embedment and deformation, in: Abu Dhabi International Petroleum Exhibition and Conference. SPE, p. D012S116R165.

Madasu, S., Nguyen, P.D., 2017. Computational and experimental study of microfracture conductivity in the eagle ford shale using microproppant pillars, in: SPE Reservoir Characterisation and Simulation Conference and Exhibition? SPE, p. D031S016R006.

Rahman, M.K., Rahman, M.M., Joarder, A.H., 2007. Analytical Production Modeling for Hydraulically Fractured Gas Reservoirs. *Petroleum Science and Technology* **25**, 683–704. https://doi.org/10.1080/10916460500411663

Warpinski, N.R., Mayerhofer, M.J., Vincent, M.C., Cipolla, C.L., Lolon, E.P., 2009. Stimulating Unconventional Reservoirs: Maximizing Network Growth While Optimizing Fracture Conductivity. *Journal of Canadian Petroleum Technology* **48**, 39–51. https://doi.org/10.2118/114173-PA